

Education

Singapore Management University, Singapore

Ph.D. in Computer Science, advised by Prof. Akshat Kumar

GPA: 4.05/4.3

Aug 2022 – Present

- SCIS Dean's List AY2026/27; Presidential Doctoral Fellowship AY2024/25, AY2025/26, AY2026/27.

University of Moratuwa, Sri Lanka

B.Sc. Eng. (Hons.) in Computer Science & Engineering, First Class

GPA: 3.99/4.2

Jan 2018

Research

Singapore Management University – Ph.D. Candidate, advised by Prof. Akshat Kumar

Aug 2022 – Present

- Focus: **Safe and constrained reinforcement learning**, **RL theory**, and generative policies for sequential decision making.
- Developed offline safe RL algorithms that learn **persistent safety** regions from fixed datasets with convergence guarantees, and **normalizing flow** and **flow-matching** policies for action-constrained control (Python, PyTorch, JAX).
- Developed RL algorithms for data-driven simulators of maritime navigation and autonomous driving.

Selected publications

full list: [Google Scholar](#)

A Choudhury*, J C Brahmanage*, A Kumar, P Paruchuri. Persistent Safety Set Guided Offline Safe Reinforcement Learning. **IJCAI 2026**

J C Brahmanage, A Kumar. Beyond Hard Constraints: Budget-Conditioned Reachability for Safe Offline Reinforcement Learning. **ICAPS 2026**

A Q Pham, J C Brahmanage, T A Mai, A Kumar. IOSTOM: Offline Imitation Learning from Observations via State Transition Occupancy Matching. **NeurIPS 2025**

J C Brahmanage, J Ling, A Kumar. Leveraging Constraint Violation Signals for Action Constrained Reinforcement Learning. (Oral) **AAAI 2025**

A Q Pham, J C Brahmanage, A Kumar. ShipNaviSim: Data-Driven Simulation for Real-World Maritime Navigation. **AAMAS 2025**

J C Brahmanage, J Ling, A Kumar. FlowPG: Action-Constrained Policy Gradient with Normalizing Flows. **NeurIPS 2023**

J C Brahmanage, T Kandappu, B Zheng. A Data-Driven Approach for Scheduling Bus Services Subject to Demand Constraints. **IEEE TKDE 2022**

* equal contribution

University of Moratuwa, Sri Lanka – Undergraduate Researcher, advised by Dr. Surangika Ranathunga

Jan 2017 – Dec 2017

- Built an annotation scheme and corpus for free-text student course feedback, and trained opinion and opinion-target extraction models to summarise it automatically.

J C Brahmanage, S Ediriweera, R Hasanth, P Munasinghe, S Ranathunga. Annotating Opinions and Opinion Targets in Student Course Feedback. **LREC 2018**

J C Brahmanage, S Ediriweera, P Munasinghe, R Hasanth, S Ranathunga. Opinion Target Extraction for Student Course Feedback. **ROCLING 2017**

Work Experience

Huawei Singapore Research Centre – Research Intern, Celia Team

May 2026 – July 2026

- Worked on **agentic search**: research and development of search and tool-use agents for Huawei's Celia assistant.

Singapore Management University – Research Engineer

Nov 2020 – July 2022

- Developed bus frequency optimisation algorithms and predicted travel times within the transport network from smart-card data; results published in IEEE TKDE with a demo system at ICDE.

Stack Technologies (Pvt) Ltd., Sri Lanka – Co-founder / Software Engineer

Oct 2019 – Nov 2021

- Led development of **Stack Scheduler**, a production planning tool for a leading shoe manufacturer; in active use six years after deployment.

Hybriteq Solutions (Pvt) Ltd., Sri Lanka – Co-founder / Software Engineer

Aug 2018 – Oct 2019

- Built mobile and web products end to end for clients.

Others: Software Engineering Intern at WSO2 2016, Google Summer of Code Contributor at Apache 2016

Other

- **Service:** Reviewer for AAAI, NeurIPS, ICAPS and JAIR. Liaison officer, Global Young Scientist Summit 2026; moderator, Singapore Rising Scholars Conference 2025.
- **Awards:** SCIS Dean's List AY2026/27; SMU Presidential Doctoral Fellowship (3×); World Finalist, NASA International Space Apps 2018 (top 5 of 25,000+).
- **Open source:** [React Native Floating Bubble](#) (218+), [Sinhala Unicode Converter](#) (19+), [ShipNaviSim](#).
- **Skills:** Python, PyTorch, JAX, NumPy/Pandas, C++, TypeScript/JavaScript, Haskell, Java.